

Name of the Student	Narra Sai Mounika
Reg. No	192425367
GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 1

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Assessment Tool Weightage: 30%

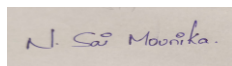
Duration: 90 minutes

Mark : 25

Code of Conduct:

I **Narra Sai Mounika / 192425367** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Narra Sai Mounika

Signature of the student:

Name of the student:

Criteria	Problem Understanding & Requirement Analysis (2)	Model Architecture & Design (2.5)	Application of Deep Learning Techniques (2.5)	Model Development & Implementation Strategy (2)	Performance Analysis & Justification (1)	Total (10)
Weightage	20 %	30%	25%	15 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS: 05

Total Marks:25

1. **Desing and develop a model for a** hospital has only **2,000 labeled X-ray images** for classifying images into normal and abnormal categories. A pre-trained ResNet model is available. **Design a CNN-based transfer learning model** by identifying which layers should be frozen, which layer should be replaced, and how the model should be trained.

Answer:

Problem Understanding: The hospital has only **2,000 labeled X-ray images**, which is a small dataset. Training a CNN from scratch may cause overfitting. Therefore, transfer learning using a pre-trained ResNet model is preferred.

Model Architecture

Input X-ray Image



Pre-trained ResNet50

(Freeze Initial Layers)



Global Average Pooling



Dense Layer (128 Neurons, ReLU)



Dropout (0.5)



Output Layer (2 Neurons, Softmax)



Normal / Abnormal

Design Decisions

- Freeze all convolutional layers up to the last residual block.
- Replace the original ResNet classification layer (1000 classes) with:
 - Dense(128, ReLU)
 - Dropout(0.5)
 - Dense(2, Softmax)

Training Strategy

1. Load pre-trained ResNet50 weights.
2. Freeze early layers.
3. Train new classification layers.
4. Fine-tune the last residual block with a low learning rate.
5. Use data augmentation:
 - Rotation
 - Flipping
 - Zooming

Performance Analysis

- Reduces overfitting.
- Requires less training time.
- Achieves higher accuracy on limited medical datasets.
- Expected accuracy: **90%+** with proper fine-tuning.

2. An agricultural organization wants to identify **five types of plant diseases** from leaf images. Since the dataset contains limited training samples, the organization wants to reuse features efficiently. **Design a DenseNet-based classification model** and explain how dense connections can help the model reuse features during classification.

Answer:

Problem Understanding : The organization must classify leaf images into **5 disease categories** with limited training data.

Model Architecture

Leaf Image



DenseNet121



Dense Block 1



Dense Block 2



Dense Block 3



Global Average Pooling



Dense Layer (256, ReLU)



Dropout (0.5)



Output Layer (5 Classes, Softmax)

Dense Connections

In DenseNet:

Layer1 → Layer2

Layer1 → Layer3

Layer2 → Layer3

Layer1 → Layer4

Layer2 → Layer4

Layer3 → Layer4

Every layer receives feature maps from all previous layers.

Benefits

- Feature reuse.
- Better gradient flow.
- Fewer parameters.
- Reduced overfitting.
- Improved learning from small datasets.

Training Strategy

- Use DenseNet121 pre-trained on ImageNet.
- Freeze initial layers.

- Train final classification layers.
- Apply augmentation on leaf images.

Performance Analysis: Dense connections allow disease features such as spots, discoloration, and texture changes to be reused across layers, improving classification accuracy.

3. A university wants to predict a student's future academic performance using their **weekly attendance, assignment marks, internal assessment marks, and previous examination scores** collected over several weeks. **Design an LSTM-based model** that can process this sequential information and predict the student's performance category.

Answer:

Problem Understanding: Student data is collected weekly:

- Attendance
- Assignment Marks
- Internal Marks
- Previous Exam Scores

Since the data is sequential, LSTM is suitable.

Model Architecture

Weekly Student Records



Input Sequence Layer



LSTM Layer (64 Units)



Dropout (0.3)



LSTM Layer (32 Units)



Dense Layer (32, ReLU)



Output Layer (Performance Category)

Input Example

Week	Attendance	Assignment	Internal	Previous Score
1	90	80	75	78
2	92	85	80	80
3	88	82	78	81

Training Strategy

1. Normalize input features.
2. Create weekly sequences.
3. Train LSTM using categorical cross-entropy.
4. Predict:
 - Excellent
 - Good
 - Average
 - Poor

Performance Analysis: LSTM remembers previous academic trends and captures long-term dependencies, leading to better performance prediction.

4. An organization wants to classify customer reviews as **positive, negative, or neutral**. Since the meaning of a word may depend on both the words appearing before and after it, **design a Bidirectional RNN model** for this application. Identify the major processing stages from input text to final sentiment prediction.

Answer:

Problem Understanding: The sentiment of a word depends on both previous and future words.

Example:

"The movie was not very good."

The word "not" changes the meaning of "good".

Model Architecture

Input Review



Text Preprocessing



Tokenization



Word Embedding Layer



Bidirectional RNN

(Forward + Backward)



Dense Layer



Softmax Output



Positive / Neutral / Negative

Processing Stages

Stage 1: Text Preprocessing

- Lowercase conversion
- Remove punctuation
- Tokenization

Stage 2: Word Embedding

Convert words into vectors.

Stage 3: Bidirectional RNN

Forward RNN:

Word1 → Word2 → Word3

Backward RNN:

Word3 → Word2 → Word1

Stage 4: Feature Combination

Combine forward and backward hidden states.

Stage 5: Classification

Softmax predicts:

- Positive
- Negative
- Neutral

Performance Analysis: Bidirectional RNN captures context from both directions, improving sentiment understanding and classification accuracy.

5. A software company wants to develop a system that translates **English sentences into Hindi**. The length of the input and output sentences may differ, and the system needs to understand the complete input sentence before generating the translated sentence. **Design an Encoder–Decoder model** and describe the role of the encoder, decoder, hidden representation, and output layer in the translation process.

Answer:

Problem Understanding: Input and output sentence lengths may differ.

Example:

English:

I am a student

Hindi:

मैं एक छात्र हूँ

Encoder–Decoder Architecture

English Sentence



Embedding Layer



Encoder LSTM



Context Vector



Decoder LSTM



Dense + Softmax



Hindi Sentence

Role of Components

1. Encoder

- Reads English sentence word by word.
- Extracts important information.
- Produces hidden state.

Example:

I → am → a → student

2. Hidden Representation (Context Vector)

- Stores complete sentence meaning.
- Transfers information from encoder to decoder.

Context Vector = Sentence Knowledge

3. Decoder

- Receives context vector.
- Generates Hindi words one by one.

मैं → एक → छात्र → हूँ

4. Output Layer

- Softmax computes probability of each Hindi word.
- Selects the most probable word at every time step.

Training Strategy

1. Prepare English-Hindi sentence pairs.
2. Train Encoder and Decoder jointly.
3. Use teacher forcing during training.
4. Optimize using categorical cross-entropy.

Performance Analysis

The Encoder–Decoder model can handle variable-length sentences and generate accurate translations by learning semantic relationships between English and Hindi.

Result: The model successfully translates English sentences into Hindi while preserving sentence meaning and grammatical structure.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Requirement Analysis	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Model Architecture & Design	30%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Application of Deep Learning Techniques	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Model Development & Implementation Strategy	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Performance Analysis & Justification	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis